

Lab 1: Film Ranking (The Basics)

Query Query History

```
1 SELECT
2     f.title,
3     c.name AS category_name,
4     f.length,
5     DENSE_RANK() OVER (PARTITION BY c.name ORDER BY f.length DESC) AS rank_len
6 FROM film f
7 JOIN film_category fc ON f.film_id = fc.film_id
8 JOIN category c ON fc.category_id = c.category_id;
9
```

	title character varying (255) 🔒	category_name character varying (25) 🔒	length smallint 🔒	rank_len bigint 🔒
1	Worst Banger	Action	185	1
2	Darn Forrester	Action	185	1
3	Casualties Encino	Action	179	2
4	Quest Mussolini	Action	177	3
5	Entrapment Satisfaction	Action	176	4
6	Lawrence Love	Action	175	5
7	Uprising Uptown	Action	174	6
8	Magnolia Forrester	Action	171	7
9	Dragon Squad	Action	170	8
10	Antitrust Tomatoes	Action	168	9
11	Campus Remember	Action	167	10
12	Speakeasy Date	Action	165	11
13	Story Side	Action	163	12
14	Fool Mockingbird	Action	158	13
15	Minds Truman	Action	149	14
16	Easy Gladiator	Action	148	15
17	South Wait	Action	143	16
18	Kissing Dolls	Action	141	17
19	Shrunk Divine	Action	139	18
20	Dream Pickup	Action	135	19
21	Sky Miracle	Action	132	20
22	Darko Dorado	Action	130	21
23	Barefoot Manchurian	Action	129	22
24	American Circus	Action	129	22
25	Gosford Donnie	Action	129	22
26	Women Dorado	Action	126	23
27	Montezuma Command	Action	126	23
28	Bull Shawshank	Action	125	24
29	Amadeus Holy	Action	113	25
30	Stagecoach Armageddon	Action	112	26
31	Celebrity Horn	Action	110	27
32	Park Citizen	Action	109	28
33	Rugrats Shakespeare	Action	109	28
34	Lord Arizona	Action	108	29

Lab 2: Growth Analysis (Time Series)

Query

Query History

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```
WITH monthly_revenue AS (  
  SELECT  
    DATE_TRUNC('month', payment_date) AS payment_month,  
    SUM(amount) AS total_revenue  
  FROM payment  
  GROUP BY 1  
)  
lagged_revenue AS (  
  SELECT  
    payment_month,  
    total_revenue,  
    LAG(total_revenue) OVER (ORDER BY payment_month) AS prev_revenue  
  FROM monthly_revenue  
)  
SELECT  
  payment_month,  
  total_revenue,  
  prev_revenue,  
  CASE  
    WHEN prev_revenue IS NULL THEN NULL  
    ELSE ((total_revenue - prev_revenue) / prev_revenue) * 100  
  END AS growth_pct  
FROM lagged_revenue;
```

Data Output

Messages

Notifications

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SQL

	payment_month timestamp without time zone 🔒	total_revenue numeric 🔒	prev_revenue numeric 🔒	growth_pct numeric 🔒
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	23886.56	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	28559.46	-98.19961581906660700200

Lab 3: Customer Lifetime Value (Running Total)

Query

Query History

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SELECT

2

customer_id,

3

payment_id,

4

amount,

5

payment_date,

6

SUM(amount) OVER (

7

PARTITION BY customer_id

8

ORDER BY payment_date, payment_id

9

) AS cumulative_clv

10

FROM payment;

Data Output

Messages

Notifications

SQL

	customer_id smallint	payment_id [PK] integer	amount numeric (5,2)	payment_date timestamp without time zone	cumulative_clv numeric	
1	1	18495	5.99	2007-02-14 23:22:38.996577	5.99	
2	1	18496	0.99	2007-02-15 16:31:19.996577	6.98	
3	1	18497	9.99	2007-02-15 19:37:12.996577	16.97	
4	1	18498	4.99	2007-02-16 13:47:23.996577	21.96	
5	1	18499	4.99	2007-02-18 07:10:14.996577	26.95	
6	1	18500	0.99	2007-02-18 12:02:25.996577	27.94	
7	1	18501	3.99	2007-02-21 04:53:11.996577	31.93	
8	1	22680	4.99	2007-03-01 07:19:30.996577	36.92	
9	1	22681	3.99	2007-03-02 14:05:18.996577	40.91	
10	1	22682	0.99	2007-03-02 16:30:04.996577	41.90	
11	1	22683	4.99	2007-03-17 11:06:20.996577	46.89	
12	1	22684	0.99	2007-03-18 02:25:55.996577	47.88	
13	1	22685	0.99	2007-03-19 08:23:42.996577	48.87	
14	1	22686	2.99	2007-03-19 12:25:20.996577	51.86	
15	1	22687	0.99	2007-03-21 22:02:23.996577	52.85	
16	1	22688	1.99	2007-03-21 23:56:23.996577	54.84	
17	1	22689	2.99	2007-03-22 18:10:03.996577	57.83	
18	1	22690	5.99	2007-03-22 18:32:12.996577	63.82	
19	1	28993	5.99	2007-04-08 01:45:31.996577	69.81	
20	1	28994	5.99	2007-04-08 06:02:22.996577	75.80	
21	1	28995	4.99	2007-04-09 11:52:33.996577	80.79	
22	1	28996	4.99	2007-04-09 15:06:27.996577	85.78	
23	1	28997	7.99	2007-04-11 08:42:12.996577	93.77	
24	1	28998	2.99	2007-04-27 09:59:48.996577	96.76	
25	1	28999	4.99	2007-04-28 07:33:11.996577	101.75	
Total rows: 14596		Query complete 00:00:00.071				

Lab 4: The "Top-N" Filter Challenge

```
WITH customer_spending AS (  
    SELECT  
        c.customer_id,  
        c.first_name,  
        c.last_name,  
        co.country,  
        SUM(p.amount) AS total_spend  
    FROM customer c  
    JOIN address a ON c.address_id = a.address_id  
    JOIN city ci ON a.city_id = ci.city_id  
    JOIN country co ON ci.country_id = co.country_id  
    JOIN payment p ON c.customer_id = p.customer_id  
    GROUP BY c.customer_id, c.first_name, c.last_name, co.country  
)  
ranked_customers AS (  
    SELECT  
        customer_id,  
        first_name,  
        last_name,  
        country,  
        total_spend,  
        RANK() OVER (PARTITION BY country ORDER BY total_spend DESC) AS spending_rank  
    FROM customer_spending  
)  
SELECT  
    customer_id,  
    first_name,  
    last_name,  
    country,  
    total_spend  
FROM ranked_customers  
WHERE spending_rank = 1;
```

	customer_id integer	first_name character varying (45)	last_name character varying (45)	country character varying (50)	total_spend numeric
1	218	Vera	Mccoy	Afghanistan	67.82
2	176	June	Carroll	Algeria	151.68
3	320	Anthony	Schwab	American Samoa	47.85
4	528	Claude	Herzog	Angola	93.80
5	381	Bobby	Boudreau	Anguilla	99.68
6	560	Jordan	Archuleta	Argentina	129.71
7	41	Stephanie	Mitchell	Armenia	118.75
8	266	Nora	Herrera	Austria	113.74
9	334	Raymond	Mcwhorter	Azerbaijan	108.75
10	590	Seth	Hannon	Bahrain	108.76
11	21	Michelle	Clark	Bangladesh	146.68
12	144	Clara	Shaw	Belarus	189.60
13	431	Joel	Francisco	Bolivia	92.78
14	178	Marion	Snyder	Brazil	194.61
15	91	Lois	Butler	Brunei	107.66
16	540	Tyrone	Asher	Bulgaria	112.76
17	516	Elmer	Noe	Cambodia	92.76
18	352	Albert	Crouse	Cameroon	96.78
19	410	Curtis	Irby	Canada	167.62
20	535	Javier	Elrod	Chad	122.72
21	45	Janet	Phillips	Chile	124.74
22	550	Guy	Brownlee	China	151.69
23	87	Wanda	Patterson	Colombia	137.72
24	375	Aaron	Selby	Congo, The Democratic Republic of ...	88.80
25	265	Jennie	Terry	Czech Republic	132.72
26	258	Myrtle	Fleming	Dominican Republic	110.76
27	494	Ramon	Choate	Ecuador	135.70
28	119	Sherry	Marshall	Egypt	139.69
29	444	Marcus	Hidalgo	Estonia	105.72
30	392	Sean	Douglass	Ethiopia	91.77
31	317	Edward	Baugh	Faroe Islands	96.76
32	298	Erika	Pena	Finland	78.79
33	35	Virginia	Green	France	118.70
Total rows: 108		Query complete 00:00:00.055			